In this type, tantalum foil acts as an anode. Using electro-chemical process, thin tantalum oxide layer is deposited on anode. This oxide layer acts as a dielectric medium. Cathode is also a tantalum foil. The total setup is placed in a sealed package.

Advantages

1. Smaller in size
2. Long term stability
3. Long shelf-life
4. Low leakage
5. Can be used in SMD form.

Disadvantages

It cannot be used in the circuits where voltage spikes are more.

Uses: It is used in

1. Cell phones.
2. Computer circuits.
3. LED Television circuits.

1.11.7.1.8 SMD Capacitor

It is directly soldered on the copper solder points of the surface of the PCB and is called as SMD capacitor. It is shown in figure 1.62

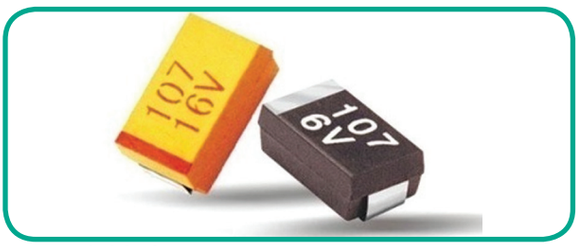


Figure 1.62 SMD capacitor

1.11.7.1.9 Ultra Capacitor

It is a type of capacitor which has higher capacitance value in farad with low voltage rating. It is an electric double layer capacitor. It is also called as super capacitors. It uses carbon as electrodes. Figure 1.63 shows an ultra-capacitor.

They can store 10 to 100 times more electrical energy than electrolytic capacitors.They can accept and deliver charge much faster than batteries, and long lasting many more charge and discharge cycles than rechargeable batteries.

Uses: It is used in DC operated cars, buses, trains, cranes and elevators and static RAM in computers.

Working Voltage

In the Figure 1.63(a), the voltage mentioned in the capacitor is the maximum voltage to be applied to a capacitor. But it

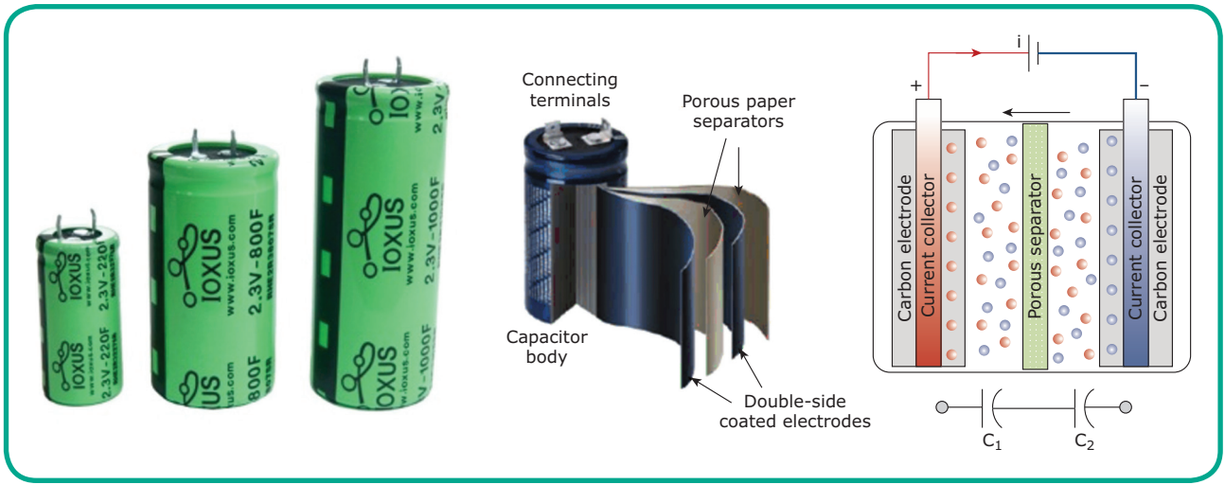


Figure 1.63 Ultra capacitor

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